TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

BMCS2063 Data Structures and Algorithms

ASSIGNMENT SPECIFICATION 202605

Introduction

Abstract data types (ADTs) are very important as they serve as programming tools that enable component reuse and encapsulation. This assignment requires students to create, implement and use ***collection ADTs*** in an application. You may do this assignment on an individual basis or in a team of up to **4 members** from the same practical class. Team assignments will normally provide a greater degree of appreciation of how various collection ADTs may be used within an application as well as the interdependence of various functionalities and features on the collection ADTs, if all members do their respective parts in a timely and responsible manner.

Learning Outcomes Assessed

CLO	Description	% to Coursework
CLO2	Produce a software program using appropriate data structures and algorithms (P4, PLO3).	85%*
CLO3	Explain the implementation and appropriateness of data structures for a specific scenario (A4, PLO5).	15%

Problem Statement

This assignment requires you to design, implement, and analyze a console-based software prototype for TARUMT Resorts, a luxury hospitality chain.

The core architectural challenge of this project is to implement a reservation and room optimization algorithms. To test your comprehensive understanding of Data Structure and Algorithms, your system must utilize both Linear and Non-Linear Abstract Data Types (ADTs) alongside explicit searching and sorting algorithms

Students are required to design and implement the system using appropriate Collection ADTs. Teams may make reasonable assumptions where necessary. Marks will be awarded based on the correctness, efficiency, creativity, and complexity of the implemented solution.

Modules involved

Modules	Descriptions
Walk-In Registrations & Standard Booking Procedure (Linear ADT)	During peak travel seasons, check-in desks handle continuous streams of guests. Standard guests must be managed in an orderly manner when room availability fluctuates due to walk-ins or unassigned rooms. Implement a Linear ADT to manage incoming standard reservations chronologically.
VIP & Loyalty Tier Priority Room Allocation (Non-Linear ADT)	High tier loyalty members (e.g.,Elite, Diamond and Platinum) bypass standard queues for room assignment. Loyalty tiers influence room assignment priority. When rooms become available, it prioritizes allocation based on membership tier rather than arrival time. The allocation structure reorganizes itself automatically upon new insertions and ensures that the guest profile with the highest tier is always positioned at the front for immediate assignment. Priority guests consistently receive first access to vacant rooms.
Housekeeping and Task Log (Linear ADT)	The housekeeping supervisor updates room cleaning statuses sequentially as staff complete tasks (e.g., Dirty → Cleaning In Progress → Inspected → Ready for Check-In). If a supervisor logs an incorrect status, or a guest requests a late check-out during cleaning, the system must instantly roll back the schedule. Room statuses are updated to reflect accurate conditions and ensure smooth coordination between housekeeping and front-desk operations.
Front-Desk Service (Non-Linear ADT & Searching)	Front-desk agents handle hundreds of simultaneous calls and walk-in inquiries. Query types involve room availability, billing details, or guest identification using a unique 8-digit confirmation number. Implement an efficient search algorithm capable of instantly retrieving complete guest information.
Loyalty and Rewards service	Manage the members' profile (personalized promotion), points accumulation and redemption, tier progression (e.g:Silver→Gold), redemption processing, notifications alert for expiring points, redemption requests, or tier upgrades.

Reports generation
- Each member is required to generate at least **two** reports for their selected module.
- At the close of every business cycle, hotel management requires analytical summaries of operational performance metrics.To fulfill this requirement, implement an algorithm that:
- Combines searching and sorting techniques
- Applies filters based on multiple criteria
- Produces a structured, well-organized console report suitable for management review and decision-making.

Scope of Work

The assignment consists of TWO components:

A. Team Component

Each team (including one-member teams for individual assignments) is required to submit the specification and implementation of **ONE** collection ADT.

Note: Although you will use a variety of collection ADTs in your assignment, each team is required to select ONLY ONE collection ADT to be submitted and assessed.

B. Individual Component

1. Each student in the team is required to develop a prototype of ONE module for the application to demonstrate your appreciation of and competency in the use of collection ADTs. There should be no duplication of subsystems among members.
   - The user interface for your application may be console-based, GUI-based or web-based. Note that no marks will be awarded for the user interface. However, the user interface should be easy to understand.
   - Since this is a prototype, the team does not need to develop the complete system/application. However, the subsystems for the team should be integrated. For the collection objects that are required for your module but are not included in any team members' scopes, you may populate the collection objects by reading from a file or using a method which adds hard-coded entity values.
   - Only validations that invoke the methods of collection ADTs are required.

2. Other than your team's collection ADT, you may also use the collection ADTs provided in the sample code for this course. However, you are not allowed to use any collection interfaces or classes from the Java Collections Framework.

Coding Standards

Teams are required to adhere to the following requirements:

1. Apply the Entity-Control-Boundary (ECB) architectural pattern to structure the classes in your NetBeans project for this assignment. The descriptions and constraints for each type of class are provided below.

   Descriptions of types of classes

   - Entity classes represent the data in the system/application. Examples of entity classes are Room, Booking, Member, others.
   - Boundary classes interact with the actors of the system. Examples of boundary classes are MaintainCourseUI, RegisterTeamUI, and GenerateReportUI.
   - Control classes implement the business logic for use cases. They orchestrate the execution of commands coming from boundary objects by interacting with entity and boundary objects. Examples of control classes are MaintainRooms, Booking, and others.
   - Utility classes (if any) contain common methods that are used by other classes to perform repetitive general tasks. A utility class contains only static methods and static variables. Thus, its methods and variables are accessed using the class name.

Constraints for each type of class

- Actors may only interact with boundary objects.
- Boundary objects may communicate with actors and control objects only.
- Control objects may communicate with boundary objects and entity objects, as well as other control objects.
- Entity objects may only know about other entity objects

You may refer to the ECBDemo NetBeans project (download the entire folder) which provides a minimal example of how to implement the ECB pattern in NetBeans.

2. All code must be written using the standard Java naming convention (i.e. using Camel Case).
3. Include your name as a comment at the beginning of each class that you authored.
4. If the code for the collection ADT was not written by you or was adapted from a source, acknowledge the source at the beginning of the Java interface class.

Assessment Criteria

Refer to the Assignment Rubrics for the assessment criteria and allocation of marks.

Assignment Deliverables and Deadlines

Prepare the team and individual assignment report using the Google Doc template provided.

Date/Week	Deliverable/Activity
Week 2: by Friday, 11.59pm **2026-06-26**	**Assignment Team Registration** One member to register your assignment team with your tutor. Please submit the Group Contract to your tutor. Get the approval from your Tutor, where & when to submit the Group Contract form. BMCS2063 Group Contract.pdf
Week 10 by Friday, 11.59pm. **2026-08-21** Submit to your tutor's Google Classroom.	**Team Assignment Report (PDF) & NetBeans project** • One member (team leader) to submit to Google Classroom**. • The NetBeans project (source code) should include any data files (e.g. text files) and also a ReadMe.txt file to explain how to run your application. • AI Usage Disclosure Form.pdf
Week 11, 12	**Assignment Demo** • Each student must demo his/her own work.

**Refer to your tutor for details and follow the naming convention required by your tutor.

Academic Integrity and Plagiarism

There must be originality in your work, i.e. do not copy or refer to other teams. You may only work with your team member(s) to produce the solution of this assignment. You must not share with or refer to any part of the assignment (including the code) of anyone else except your team member(s) and your tutor.

Before submitting your assignment, ensure that you have complied with TAR UMT's plagiarism policy. Any cheating, attempt to cheat, plagiarism, collusion and any other attempts to gain an unfair advantage in assessment will cause the students concerned to be penalized. Students found to be dishonest are liable to disciplinary action.

AI Use in Assessment: Yellow (Limited AI)
AI tools may be used for specific and clearly defined purposes, such as data analysis, language refinement, or technical support. However, AI tools shall not be used to generate core arguments, critical analysis, or original academic content.

AI Usage Guidelines: AI tools may be used for foundational support, including explaining data structures concepts, or checking syntax help. Students are not permitted to use AI to generate entire application source codes, draft assignments, or complete primary modules or problem sets.

Transparency and Accountability: Seek clarification from the lecturer or tutor before submission. Any unauthorised use of AI to generate core content will be considered a violation of academic integrity policies.

Students are required to disclose AI usage using this form.

AI Usage Disclosure Form.pdf

Late Submission

Students are required to submit the Late Submission of Coursework Form The together with their assignment.

Refer to TAR UMT's Procedure Relating to Coursework Submission for more details.